

Step 1: Data Loading and Understanding

1.1 Datasets Used

Two datasets were used. The ACN dataset contains session-level EV charging records from Caltech/JPL sites in the US, comprising 16,304 sessions across 27 fields. It was provided as an Excel export of the original JSON (acndata_sessions.json.xlsx) and is reserved for session-level behavioral EDA and cross-context pricing validation. The UrbanEV (ST-EVCDP) dataset contains zone-level operational data for Shenzhen, China — volume, occupancy, duration, and price — recorded at 5-minute intervals over a 30-day period (19 June to 18 July 2022). UrbanEV serves as the primary source for the dynamic-pricing modeling pipeline because it contains real observed price variation across zones and over time, a property the ACN data lacks (ACN records charging sessions but carries no per-session tariff field).

1.2 Data Loading

The datasets were loaded into Python using pandas. The ACN sessions file was loaded via `read_excel`. The following UrbanEV files were loaded into the core pipeline: `volume.csv`, `price.csv`, `occupancy.csv`, `duration.csv`, `time.csv`, and `information.csv`. `stations.csv` was loaded and inspected but not integrated: it describes 1,706 individual stations, whereas the operational matrices are aggregated to 247 traffic zones, a finer granularity with no clean one-to-one mapping. To avoid an unreliable join, `information.csv` (247 zones) was used as the authoritative zone-level metadata source. The spatial files `adj.csv` (adjacency matrix) and `distance.csv` (distance matrix) were examined and excluded, as they support graph-based spatial models (e.g. GCN/GAT) that were outside the scope of this implementation.

1.3 Structure and Quality Inspection

The shape of every file was verified. The operational files are wide matrices of shape $(8,640 \times 248)$: one row per 5-minute timestamp and one column per traffic zone, plus a leading timestamp column (8,640

timestamps × 247 zones + 1 timestamp column). Because they are wide rather than tidy, a reshape to long format (one row per timestamp-zone pair) is required before integration; this is carried out in preprocessing. The volume, price, occupancy, and duration matrices share identical dimensions, confirming they can be aligned cell-for-cell via the shared timestamp index and zone columns. A null-value check returned zero missing values across all four operational matrices. In the ACN preview, several columns (`_meta`, `end`, `min_kWh`, `start`, `_items`) were entirely empty — JSON metadata artifacts to be dropped during preprocessing.

1.4 Timestamp Construction

UrbanEV stores time as separate year, month, day, hour, minute, and second columns. These were combined into a single datetime feature. This confirmed the data spans 19 June 2022 00:00 to 18 July 2022 23:55 at a consistent 5-minute interval across all 8,640 rows (30 days).

1.5 Metadata Join Key

When merging zone metadata from `information.csv` onto the operational matrices, the join key is the grid column (zone codes 102, 105, 107, ...), which matches the matrix column headers — not the num column (an internal index 1, 3, 5, ...). A spot-check confirmed alignment: grid 102 maps to a total charger count of 30 (3 fast + 27 slow), and the leading grid codes (102, 105, 107, 108, 109) match the matrix column headers exactly.

1.6 Confirmation of Price Variation

Because UrbanEV's value to the project rests on it containing genuine price variation, this was verified directly. 57 of the 247 zones are flagged as dynamic-pricing zones in the metadata. The price matrix ranges from 0.25 to 1.47 per kWh (mean ≈ 0.96), with 210 distinct zone-level mean prices and 240 of 247 zones showing price changes over time. This confirms that UrbanEV captures real cross-zone and

temporal price variation, justifying its selection as the primary dataset for the pricing pipeline.

1.7 Key Observations

UrbanEV contains 8,640 timestamps at 5-minute granularity over 30 days. The volume, occupancy, duration, and price matrices share identical dimensions and align via the timestamp index and zone columns. Zone-level metadata (charger count, area, location, CBD flag, dynamic-pricing flag) is available in information.csv for segmentation in EDA. stations.csv was excluded due to a granularity mismatch with the zone-level matrices, and the spatial adjacency/distance matrices were excluded as they pertain to graph-based methods outside this scope.

Step 2: Cleaning & Alignment

Two datasets were prepared at this stage. They were cleaned together for organizational clarity but were never merged, as they represent different networks serving different roles: UrbanEV (Shenzhen, zone-level) was integrated into a single unified analytical base for demand modeling and pricing, while ACN (US, session-level) was cleaned into a separate session table used later to evaluate the pricing and monitoring agents.

The brief calls for alignment by timestamp, station ID, and session granularity into a unified analytical base. These descriptors map onto two structurally different datasets that share no common join key: UrbanEV is zone-level interval data (Shenzhen), while ACN is session-level data from a separate network (US workplace sites). The two cannot be meaningfully merged into one table, as they describe different cities, different infrastructure, and different units of observation. Accordingly, a single unified analytical base was built from UrbanEV at zone-timestamp granularity (the primary modeling

dataset, aligned by timestamp and zone), while ACN was aligned at its own session granularity as a separate table. This separation is deliberate: each dataset is prepared at the granularity native to its role, and neither is forced into an artificial join that would compromise data integrity.

2.A UrbanEV — Unified Analytical Base

2.A.1 Timestamp alignment. The operational matrices index time as a sequential integer (1, 2, 3...) representing row order rather than clock time. A matching integer timestamp key was created on the time table so the real datetime built in Step 1 could be joined onto every record, anchoring all 8,640 intervals to real dates from 19 June to 18 July 2022 at 5-minute granularity.

2.A.2 Reshaping (wide → long). Each operational matrix is stored in wide form (8,640 timestamps × 247 zone columns), which is unsuitable for merging or analysis. Each was reshaped to long format via a melt operation, producing one row per timestamp-zone pair (2,134,080 rows). This tidy structure is the prerequisite for integration and feature engineering.

2.A.3 Integration of operational data. The four reshaped tables — volume, occupancy, price, and duration — share identical (timestamp, zone) keys and were merged on both keys into a single table of 2,134,080 rows. A shape assertion confirmed the merge introduced no row fan-out from duplicated keys.

2.A.4 Attaching real time. The real datetime was joined onto the unified table via the timestamp key, enabling later analysis by hour-of-day, day-of-week, and weekday versus weekend.

2.A.5 Attaching zone metadata. Zone-level metadata from information.csv — charging capacity, fast/slow pile counts, area, coordinates, CBD flag, and dynamic-pricing flag — was merged onto the table. The join used the grid field (zone codes 102, 105, ...), which matches the matrix column headers, rather than the num field, which

is an unrelated internal index. Zone codes were cast to string on both sides for exact matching, and the count field was renamed to capacity to reflect its role as the denominator of the utilization rate. After this merge the unified base had 15 columns.

2.A.6 Merge validation. A spot-check confirmed the merge: zone 102 returned a capacity of 30, matching information.csv, and zero rows had missing capacity, confirming all 247 zones matched their metadata. Because a mismatched key fails silently and yields blank capacities rather than an error, this check was essential.

2.A.7 Completeness. A null-value check across all columns returned zero missing entries, confirming the operational matrices and metadata were fully populated after the merge.

2.A.8 Validity and duplicate checks. Checks for logically impossible values found no negative volume, no negative occupancy, and no zero or negative prices. A check for duplicate (timestamp, zone) pairs returned zero, confirming each of the 2,134,080 records is a unique zone-timestamp observation.

2.A.9 Occupancy–capacity correction. A check for records where occupancy exceeded listed capacity identified 159 records (0.007% of the data), all from a single zone (715), each showing occupancy of 14 against a capacity of 13 — a uniform off-by-one mismatch of approximately 7.7%. The uniformity of the discrepancy (identical in every affected row, confined to one zone) indicates a minor inconsistency between the fixed capacity snapshot in information.csv and actual pile availability, rather than genuine over-utilization. To keep the downstream utilization rate bounded at 100%, a new occupancy_capped field was created by capping occupancy at capacity; the original occupancy values were retained unchanged for transparency. This brought the final unified base to 2,134,080 records across 16 columns.

Temporal resolution note. Inspection of consecutive records showed that operational readings often repeat across consecutive 5-minute intervals, indicating the source data is forward-filled between less-frequent true updates. This is a characteristic of the dataset rather than an error, and implies genuine temporal variation occurs at a coarser interval than 5 minutes — relevant when interpreting short-run volatility.

2.B ACN — Session Cleaning

ACN is a different network (US workplace sites) from UrbanEV and was never merged with the unified base; it was cleaned here only for organizational convenience. Its role is downstream, supplying the session-level energy and timing used to evaluate the Tariff Pricing and Monitoring agents (Revenue Gain %, Customer Response Rate, Pricing Efficiency).

2.B.1 Column selection. Of ACN's 27 raw fields, most were either empty artifacts of the JSON-to-tabular conversion or sparsely populated optional user-input fields (over 78% missing). Only the reliably populated core session fields were retained: sessionID, stationID, spaceID, clusterID, the three timestamps (connectionTime, disconnectTime, doneChargingTime), and kWhDelivered. Rows missing essential fields were dropped, leaving 14,999 valid sessions.

2.B.2 Feature engineering. The timestamps, stored as strings (e.g. "Wed, 25 Apr 2018 11:08:04 GMT"), were parsed to datetime. Session duration (plug-in time) and actual charging time were derived in hours, along with time-of-day features (hour, day-of-week, weekend flag) from the session start.

2.B.3 Validity checks and corrections. No sessions had non-positive duration or zero energy, so none were removed on those grounds. Eighteen sessions (0.12%) showed a negative charging time, where doneChargingTime was recorded before connectionTime — a minor source glitch. Rather than drop these otherwise-valid sessions, their

charging-time values were set to missing, preserving the intact energy and duration fields that the downstream revenue metrics rely on. Separately, a small number of sessions showed extreme plug-in durations (one exceeding 240 hours). To prevent these from distorting duration-based summaries, session length was capped at the 99th percentile, a documented and conservative cap affecting only the longest ~1% of sessions; energy delivered was left unchanged.

2.B.4 Summary. The cleaned ACN table spans 25 April to 16 December 2018 (approximately eight months), comprising 14,999 sessions and 135,028 kWh of total energy delivered, averaging 9.0 kWh per session. After nulling the 18 negative charging-time glitches and capping the longest 1% of durations at 24 hours, the mean session length is 5.68 hours — all consistent with workplace charging behavior.

Step 3: Feature Engineering

Raw operational measurements on the unified UrbanEV base were translated into economically meaningful signals for demand prediction and pricing. ACN's session-level features were created earlier during cleaning (Step 2.B); the utilization, congestion, and density features below are zone-interval constructs with no session-level equivalent, so they apply to UrbanEV only. All prices and derived margins are expressed in the UrbanEV price unit (¥/kWh).

3.1 Temporal features. The datetime field was decomposed into hour-of-day, day-of-week, calendar date, and a weekend indicator. These enable analysis and prediction of demand across intraday cycles and weekday-versus-weekend patterns, central to time-of-day pricing.

3.2 Charger utilization rate. Two measures were computed. The time-based measure, following the brief's definition (charging time ÷ total available time), is duration divided by available pile-hours, where

available pile-hours equals zone capacity multiplied by the 5-minute interval (1/12 hour). It was capped at 100%, though only 50 of 2.13 million rows (0.002%) required capping, confirming the cap is cosmetic rather than masking a problem. Its mean is approximately 0.226, indicating chargers operate on average at roughly a quarter of capacity. A second, occupancy-based measure (busy piles ÷ total piles) was also computed, with a mean of approximately 0.280; it provides a cleaner signal for the surge/discount bands. The two are strongly correlated ($r \approx 0.85$), confirming a consistent story. The time-based measure is reported as the brief's primary definition, while the occupancy-based measure is used operationally for the pricing logic.

3.3 Revenue and revenue per session. Interval revenue was computed as energy delivered multiplied by price. Revenue per active pile was derived as interval revenue divided by occupancy (busy piles), excluding the roughly 49,590 zero-occupancy intervals to avoid division by zero. Because occupancy counts concurrently busy piles rather than discrete sessions, this measure is best read as revenue per active pile-interval rather than strictly per session.

3.4 Energy cost per kWh. The dataset contains the selling price but not the operator's procurement cost. Procurement cost was therefore approximated as a fixed 60% of selling price, with a per-kWh margin derived as price minus this cost. This 60% ratio is an explicit assumption documented in the absence of cost data. The resulting margin is positive across all records (minimum ≈ 0.10 ¥/kWh), confirming the assumption produces no loss-making prices.

3.5 Queue length proxy. As the dataset records only active charging and contains no direct count of waiting vehicles, congestion was estimated indirectly. A zone at or above 80% utilization was treated as saturated, on the basis that demand arriving during saturation must wait. The queue proxy was defined as utilization above the 80% threshold scaled by zone capacity, with a binary congestion flag

alongside. The 80% threshold aligns with the brief's surge-pricing trigger, giving a single consistent definition of saturation. Only about 1.07% of intervals are congested under this definition.

3.6 Occupancy density. Two density measures were created: a raw measure (occupancy relative to capacity) capturing absolute fullness, and a relative measure (utilization relative to each zone's own average) capturing whether a zone is busier than normal for itself. The relative measure averages to exactly 1.0 by construction, with no infinite or missing values, providing a normalized way to compare activity across zones of differing baseline demand.

3.7 Pricing bands. Each interval was classified by the occupancy-based utilization into a band aligned with the brief's pricing rules: high (above 80%, a surge candidate), low (below 30%, a discount candidate), and medium (in between, price held). This labelling is the direct interface between the demand signal and the downstream Tariff Pricing Agent.

3.8 Key finding from feature distributions. Examination of the bands revealed the defining characteristic of the network: 60.9% of all intervals fall in the low-utilization band (below 30%), 38.1% in medium, and only 1.0% reach the 80% saturation threshold (with just 0.35% above 90%). The network is therefore predominantly under-utilized rather than congestion-bound. This directly shapes the pricing strategy: the dominant lever is off-peak discounting to stimulate latent demand and improve utilization, while surge pricing remains a minor, targeted tool for the rare saturated intervals. This interpretation carries forward into the exploratory analysis and pricing model design.

Assumptions and limitations. Energy cost is assumed at 60% of selling price, as the dataset provides no procurement-cost field; reported margins and any profit-based conclusions are sensitive to this assumption. Two utilization definitions were computed — time-based (the brief's wording) and occupancy-based (used for pricing bands) —

because each captures a slightly different aspect of charger use; they are strongly correlated ($r \approx 0.85$).

Missing-Value Handling and Assumptions Log

In keeping with the brief's requirement for transparent, documented handling at each stage, all missing-value decisions and modeling assumptions made during preprocessing and feature engineering are consolidated here.

Missing-value handling. The UrbanEV operational matrices (volume, occupancy, price, duration) and zone metadata contained zero missing values, verified by an explicit null check after merging (Step 2.A.7); no imputation was required for this dataset. In ACN, 1,305 rows were empty artifacts of the JSON-to-tabular conversion and were dropped on the essential session fields (Step 2.B.1), leaving 14,999 valid sessions. No further missingness remained in the retained ACN fields.

Data-quality corrections. A single zone (715) showed occupancy exceeding its listed capacity in 159 records (a uniform off-by-one of $\sim 7.7\%$), attributed to a fixed capacity snapshot rather than genuine over-utilization; occupancy was capped at capacity in a new field, with the original values retained for transparency (Step 2.A.9). For revenue-per-session, approximately 49,590 zero-occupancy intervals were excluded from that single feature to avoid division by zero (Step 3.3); these intervals remain in the base for all other purposes.

Documented assumptions. Operator procurement cost is assumed at 60% of selling price, as the dataset contains no cost field; all margin and profit-based figures are sensitive to this assumption (Step 3.4). The saturation/congestion threshold is set at 80% utilization to align with the brief's surge-pricing trigger, giving a single consistent definition across the analysis (Step 3.5). Two utilization definitions are maintained — time-based (the brief's wording) and occupancy-based (used for pricing bands) — as each captures a distinct aspect of charger use; they are strongly correlated ($r \approx 0.85$).

Step 4: Exploratory Data Analysis

All visualizations were designed to be insight-driven, well-labeled, and tied directly to pricing implications. Utilization throughout refers to the occupancy-based measure. The analysis progresses from overall utilization structure through temporal, spatial, and station-type patterns, to volatility quantification.

4.1 Utilization distribution. Classifying all intervals into bands showed 60.9% fall below 30% utilization (low), 38.1% between 30–80% (medium), and only 1.0% exceed 80% (high). The network is predominantly under-utilized and only rarely congested. Pricing implication: the dominant lever is off-peak discounting to stimulate latent demand, with surge pricing relevant only for the rare high-utilization intervals.

4.2 Intraday demand cycle. Average utilization by hour revealed an inverted profile: highest overnight (peaking at hour 1, ~0.334) and lowest in the daytime working hours (trough at hour 16, ~0.232). Utilization never approaches the 80% surge threshold at any hour. This is consistent with a fleet-oriented network rather than private commuters. Because UrbanEV carries no user-type label, fleet-versus-public usage is inferred behaviorally: the overnight-peaked, daytime-trough profile is the signature of a fleet-dominated network (e.g. electric taxis or buses that operate during the day and charge while idle overnight). Pricing implication: the daytime block is a large, predictable low-demand window suited to discount pricing; surge pricing has limited applicability since demand never nears saturation. (Note: individual zone-intervals do reach saturation — the 1% high band — even though hourly network averages do not, which is why surge logic is retained as a targeted tool.)

4.3 Weekday vs weekend. Weekday and weekend profiles were nearly identical (means 0.279 and 0.283), with no pronounced weekend decline and only a slight weekend daytime uplift. The absence of a weekend drop reinforces the fleet-dominated reading, as a network serving private commuters would show a clear weekend fall. Pricing implication: the daytime discount window is consistent every day, allowing a stable, always-on off-peak rule rather than day-type-dependent logic.

4.4 Demand heatmap. A heatmap of utilization across hour and day-of-week visually confirmed the consistent overnight-high, daytime-low pattern across all seven days, with no distinct weekday or weekend signature.

4.5 Zone-level utilization. Ranking zones by mean utilization spanned from 0.062 (most under-utilized) to 0.76 (most utilized). The majority of zones operate at low average utilization, with only a small number sustaining high usage. Pricing implication: under-utilized zones are primary discount targets; the few high-demand zones are candidates for selective premium pricing.

4.6 Price–demand relationship. Mean utilization declines across price deciles, with an overall correlation of -0.28 , consistent with negative price elasticity of demand. This is an observational association rather than a causal estimate — price and demand may be jointly influenced by location and time — and is used only as directional support for the elasticity assumption. The within-segment check (4.7) further qualifies it.

4.7 CBD vs non-CBD. Business-district (CBD) zones show a flat profile throughout the day, while non-CBD zones show the pronounced overnight peak and daytime trough (mean 0.293 vs 0.243). Critically, the price–demand relationship differs sharply by zone type: non-CBD zones show clear elasticity (correlation -0.30), while CBD zones show negligible price response (-0.05). This non-CBD correlation of -0.30 is

the empirical basis for the -0.3 elasticity assumption used in the pricing model. Pricing implication: dynamic pricing will effectively shift demand in fleet-oriented non-CBD areas but is largely ineffective in price-insensitive CBD zones — strong evidence for a zone-differentiated strategy rather than a uniform tariff.

4.8 Revenue by hour. Average interval revenue mirrors the utilization curve — highest overnight and lowest during mid-morning (09:00–11:00), with a secondary bump around midday. Pricing implication: the daytime revenue trough quantifies the size of the discount opportunity, where attracting currently-absent demand could lift revenue from a low base.

4.9 Volatility across periods. Volatility was quantified by standard deviation and coefficient of variation per period: peak (mean 0.326, std 0.196, CV 0.600), shoulder (0.287, 0.173, 0.603), and off-peak (0.244, 0.152, 0.625). Peak periods carry the highest absolute volatility but the lowest relative volatility, while off-peak periods are the most relatively unpredictable. CV is substantial (~ 0.60) and fairly uniform across periods. Pricing implication: demand is volatile across all periods, justifying dynamic pricing broadly rather than in a single window; the slightly higher off-peak CV indicates greater uncertainty in the discount-period demand the model must predict.

4.10 Long-run trend. Daily average utilization rose modestly over the 30-day period (first five days 0.272 vs last five days 0.292), indicating gradually increasing usage, with a brief dip in early July. Pricing implication: the network is on a mild growth trajectory, providing a stable base. Modeling note: because demand trends upward over time, the train/test split in the prediction stage must be time-ordered rather than random, to avoid the model learning from future data.

4.11 Fast vs slow charger zones. Fast-majority zones show a far more pronounced and variable overnight pattern (mean 0.382, peaking ~ 0.52 overnight), while slow-majority zones are flatter and lower

(mean 0.268). Pricing implication: fast-charger zones — high demand, high variability, deep daytime troughs — offer the greatest leverage for dynamic pricing, whereas flat slow-charger zones respond less to time-based pricing.

4.12 Spatial distribution. Mapping mean utilization across Shenzhen showed under-utilization is geographically widespread rather than localized; most zones, including the dense central cluster, operate at low utilization, while the few high-demand zones are spatially scattered. Pricing implication: this supports a network-wide off-peak discount strategy to address systemic demand shortfall, complemented by selective zone-level premium pricing for isolated high-demand sites. Limitation: without an underlying street map, zone roles are inferred from the CBD and charger-type analyses rather than read directly from location.

Summary of EDA findings. The analysis established a coherent, evidence-backed picture: a predominantly under-utilized network (4.1) with an overnight-peaked, fleet-style demand pattern (4.2) operating seven days a week (4.3); price elasticity concentrated in non-CBD, fast-charger zones (4.6, 4.7, 4.11); widespread spatial under-utilization (4.12); and substantial demand volatility across all periods (4.9). Together these motivate a zone-differentiated, discount-led dynamic pricing strategy — stimulating demand in the large daytime trough where price-sensitive fleet users dominate, while applying surge pricing selectively to the rare saturated intervals.

Step 5: Demand Prediction Agent

5.1 Objective and design. The Demand Prediction Agent forecasts near-term charging demand to inform pricing decisions. Per the specification, it produces three outputs: predicted utilization rate, congestion probability, and expected charging load. As these are

statistically distinct problems — two regression tasks on different scales and one rare-event classification — each was modeled separately with the most suitable algorithm, rather than forcing one model to produce heterogeneous outputs. All models were trained on the UrbanEV dataset, per the dataset-to-deliverable mapping. The brief's reference to "historical session features" is operationalized here as historical zone-interval demand features, since UrbanEV — the designated modeling dataset — is interval-level rather than session-level.

5.2 Data preparation. The 5-minute records were aggregated to an hourly grain per zone, reducing data volume and mitigating the forward-fill repetition present at finer granularity while preserving the daily demand cycle. Aggregation was applied per variable: utilization and price were averaged across the twelve intervals in each hour, hourly charging volume was averaged to represent typical load, and static zone attributes (capacity, CBD flag, charger counts) took the first value. This produced 177,840 zone-hour records. For each zone, lag and rolling features were engineered: utilization at 1, 2, 3, and 24 hours prior, and a 3-hour rolling mean, with equivalent lags for charging volume. All features were constructed strictly from past values (the rolling mean was shifted before averaging), sorted by zone and time, to prevent leakage across zones or the train/test boundary. Records lacking sufficient history were removed, leaving 171,912 rows.

5.3 Train/test split. A chronological 80/20 split was used (cutoff 13 July 2022), with the earliest 80% for training (137,579 rows) and the most recent 20% for testing (34,333 rows). A random split was deliberately avoided because the data is a time series with an upward demand trend (Section 4.10); shuffling would leak future information and inflate performance. The chronological split mirrors real deployment, where the model predicts a future it has not observed.

5.4 Utilization rate prediction. Four approaches were compared on the test set: naive persistence (RMSE 0.0545, R^2 0.902), Linear Regression (0.0478, 0.925), Random Forest (0.0423, 0.941), and XGBoost (0.0391, 0.950). XGBoost performed best and was selected, improving on Random Forest by approximately 8% and on the naive baseline by approximately 28% in RMSE. The naive baseline was included specifically to test whether the model adds value beyond simple persistence; the clear improvement confirms it captures genuine demand dynamics — particularly at transition hours where demand is rising or falling — rather than merely exploiting hour-to-hour autocorrelation.

5.5 Congestion probability prediction. Congestion was defined as an hour in which zone utilization reaches or exceeds 80%. Because such events are rare — only 0.94% of hours at the hourly grain — a class-balanced Random Forest classifier was used to prevent the model defaulting to an always-"not congested" prediction. The classifier achieved an ROC-AUC of 0.997, with a predicted congestion rate (1.1%) closely tracking the actual test-set rate (0.9%), indicating it discriminates congested intervals reliably rather than exploiting class imbalance. ROC-AUC is reported as the primary metric, as raw accuracy (99.5%) is uninformative given the rarity of congestion.

5.6 Charging load prediction. Expected charging load (hourly energy volume per zone) was modeled with Random Forest, achieving R^2 0.974 (RMSE 17.05, MAE 5.36). The relatively large RMSE reflects the wide range of energy volumes across zones (standard deviation ~ 101); the high R^2 and comparatively small MAE confirm the model tracks load accurately across that range.

5.7 Feature importance and interpretation. For the selected XGBoost model, predictive importance was distributed across recent demand history rather than concentrated in one feature: previous-hour utilization (0.56), the 3-hour rolling mean (0.31), and same-hour-

yesterday utilization (0.07) together accounted for the large majority (0.94) of importance, with calendar and zone features (charger type, hour, CBD, capacity) contributing modest refinements. This indicates the model captures both the current demand level and its short-term direction, explaining its accuracy advantage at transition hours over pure persistence. Demand is therefore characterized as highly autocorrelated and short-horizon predictable; calendar and zone effects add value primarily once recent demand is known. The strength of this autocorrelation is partly amplified by the forward-filled nature of the source data.

5.8 Agent output. The agent's combined forecast — predicted utilization rate, congestion probability, and expected charging load per zone per hour — was assembled for the held-out test period (34,333 rows) and exported as `demand_forecast.csv` for the Tariff Pricing Agent. Utilization predictions were clipped to the valid 0–1 range.

Summary. The Demand Prediction Agent delivers all three required outputs with strong, honestly-validated performance: utilization (XGBoost, R^2 0.95, a 28% RMSE improvement over naive persistence), congestion probability (ROC-AUC 0.997), and charging load (R^2 0.97). The methodology emphasizes leakage-free temporal features, a deployment-realistic chronological split, and benchmarking against a naive baseline to demonstrate genuine predictive value.

Assumptions and limitations. The model is a short-horizon (one-hour-ahead) forecaster whose accuracy depends heavily on recent demand history; performance over longer horizons, where recent lags are unavailable, would be lower and was not evaluated. The strong autocorrelation exploited by the model is partly amplified by forward-filled source data, so reported accuracy may overstate performance on a system with true 5-minute variation.

Step 6: Tariff Pricing Agent

6.1 Purpose. The Tariff Pricing Agent takes the demand forecast from Step 5 and converts it into a recommended price for each zone and hour, charging more when demand is high relative to the network's norm and less when low, to balance usage across the day while keeping prices acceptable. Revenue was evaluated on the ACN sessions against the project's flat ₹15/kWh baseline. The dynamic price is a dimensionless multiplier applied to the ₹15 baseline, so all revenue figures are denominated in ₹; UrbanEV's own price units do not enter the revenue calculation.

6.2 Two pricing strategies. Following the company's guidance that the fixed 80%/30% thresholds could be optimised, two strategies were built and compared. The threshold strategy is the brief's rule-based approach: surcharge the network's busiest hours, discount the quietest, hold at baseline otherwise. The smooth strategy is an optimised continuous function where price scales gradually with predicted demand, avoiding sharp jumps. An important design correction was made: an initial version pivoted prices around a fixed 50% utilization point, but because this network operates far below that level (mean predicted utilization 0.288), that caused every hour to be discounted. Both functions were therefore re-centred on the network's actual mean utilization (0.288), so hours busy relative to this network's normal level receive a surcharge and genuinely quiet hours a discount. The network-relative thresholds resolved to a surcharge above 0.42 utilization and a discount below 0.18. After this correction, prices ranged both above and below baseline — higher during the busy overnight period, lower during the quiet daytime — consistent with the EDA demand patterns. Here "surge" denotes pricing up relatively-busy hours, not absolute congestion at or above 80%, which is rare.

6.3 Demand-response assumption. Because the historical data records demand only at the prices actually charged, an assumption is required for how demand reacts to new prices. A standard constant-elasticity rule was used: a price rise reduces demand proportionally, and vice versa. As the true elasticity is unknown, three values were tested (-0.2 , -0.3 , -0.5), with -0.3 as the central estimate grounded in the -0.30 non-CBD price–demand association observed in EDA (4.7), bracketed by -0.2 and -0.5 for sensitivity.

6.4 Consumer-acceptability bounds. All prices were capped within a bound of the baseline (default $\pm 30\%$). To understand the trade-off, the bound was swept across $\pm 10\%$, $\pm 20\%$, $\pm 30\%$, and $\pm 50\%$ to measure how revenue responds as wider swings are permitted.

6.5 Results. The optimised smooth strategy consistently outperformed the rule-based threshold strategy at every elasticity level (smooth -0.31% to -0.45% ; threshold -2.23% to -3.33% revenue change versus baseline), validating the decision to optimise the pricing function rather than rely on fixed thresholds. Revenue change was marginally negative and very close to break-even (within roughly 0.5% of baseline for the smooth strategy). This is an honest and important finding: because the network is mostly under-utilized and demand responds only weakly to price, dynamic pricing here is not a revenue-growth tool — its value lies in redistributing demand toward underused periods at negligible revenue cost. Higher assumed elasticity improved results (-0.5 gave -0.31% versus -0.45% at -0.2), because stronger demand response means discounts attract more compensating volume. The price distribution explains the strategy gap: smooth prices stay tightly around baseline (mean ₹15.00, interquartile range ₹13.88–15.86), whereas the threshold strategy jumps to the bound extremes (₹10.50 or ₹19.50), making it blunter. The hourly price profile confirms the intended pattern — prices are highest overnight (around ₹15.5 at hours 0–6) and lowest in the

daytime (around ₹14.5 at hours 11–17), mirroring the fleet demand curve.

6.6 Price-bound sensitivity. Revenue change stayed within a narrow band as the bound widened from $\pm 10\%$ to $\pm 50\%$ — from -0.84% to -0.38% , a spread of under half a percentage point. Widening the cap yields only marginal revenue change, because the smooth prices rarely approach the bound extremes. This supports the company's concern about drastic pricing: since wider price swings yield negligible revenue benefit, there is no revenue justification for imposing large, consumer-unfriendly changes, and conservative bounds can be used at near-zero cost.

6.7 Conclusion. The agent produces sensible, network-aware dynamic prices that surcharge relatively-busy periods and discount quiet ones. The pricing strength was set to 0.6 — the reward-maximising intensity identified by the Monitoring Agent's learning loop (Step 7) — so the deployed prices reflect the learned-optimal setting. The key business takeaway is that in a low-utilization, price-inelastic network, dynamic pricing should be positioned as a demand-balancing and utilization-improvement tool operating at near-revenue-neutrality, rather than a revenue-maximisation mechanism. The full benefit — increased usage during off-peak periods — is quantified by the Monitoring Agent in the next step.

Assumptions and limitations. The elasticity rule is an assumption, not a measured value; revenue results depend on it, which is why a range (-0.2 to -0.5) was tested rather than a single figure. The price-reaction steepness parameter was set to 0.6, the value found reward-optimal by the Step 7 learning loop; results are robust to it given the demonstrated insensitivity to bound width. Dynamic prices were derived from UrbanEV demand patterns and applied to ACN sessions (different networks); this is a transferability test consistent with the

project's dataset mapping, not a same-network evaluation, and results should be read in that light.

Step 7: Monitoring & Learning Agent

7.1 Purpose. The Monitoring & Learning Agent is the evaluation and feedback layer. It checks whether the Tariff Agent's pricing decisions achieved their goals using three operational metrics, then runs a learning loop to identify the best pricing intensity. Per the dataset mapping, the wait-time metric uses UrbanEV as a proxy, while customer response and pricing efficiency use ACN with the elasticity assumption. The three metrics are evaluated at a pricing strength of 0.6 — the reward-maximising intensity identified by the learning loop below — so the reported metrics reflect the system's optimal configuration.

7.2 Waiting-time reduction (proxy). With no direct queue measure, congestion served as the proxy: the share of intervals at or above 80% utilization, where additional arrivals would wait. Baseline congestion was measured on the model's predicted utilization for the test period (0.65%) — lower than the actual rate (0.94%) because the model smooths extremes — and compared against the demand resulting once dynamic prices shift it. Under dynamic pricing, the surcharge on busy periods reduced congested intervals from 0.65% to 0.30%, a relative reduction of 54.46%. This is a large relative improvement but should be read in context: congestion was already rare in this under-utilized network, so the absolute level of waiting was low to begin with. The pricing nonetheless meaningfully smoothed the limited congestion that did occur.

7.3 Customer response rate. This measures how much charging demand shifts in response to the new prices, estimated via the elasticity assumption. Total energy demand rose by +0.16% under dynamic pricing (135,028 to 135,248 kWh), with +0.45% of demand

moving into the cheaper discounted hours — the off-peak uplift. These shifts are positive but modest, consistent with the network's low price-sensitivity found in EDA and the gentle, consumer-acceptable pricing intensity selected: demand responds in the expected direction, but only slightly. The positive off-peak uplift confirms the pricing achieves its core intended effect — pulling demand toward underused periods.

7.4 Pricing efficiency score. Defined as revenue per kWh delivered, this tracks whether pricing improves the economic value of each unit of energy. Under dynamic pricing, efficiency was ₹14.92/kWh versus the ₹15.00 baseline (−0.56%). The very slight decrease reflects the discount-led nature of pricing in a low-utilization network and is consistent with the near-break-even revenue result from Step 6.

7.5 The learning mechanism. To make the system self-improving rather than static, the agent evaluates a range of pricing intensities ("strength", from 0.1 to 1.2) against a combined reward that credits congestion reduction and penalises revenue loss. The reward landscape reveals a clear structure: congestion reduction saturates at strength 0.5 (the congested-interval rate bottoms out at 0.30% and falls no further beyond that point), while revenue loss continues to grow steadily with strength (from −0.32% at 0.5 to −1.52% at 1.2). The reward therefore rises steeply to a broad, flat plateau across strengths 0.5–0.6, peaking at 0.6 (reward 54.06), then declines slowly as additional intensity sacrifices revenue without further congestion benefit. The agent selects strength 0.6 as the reward-maximising setting, and this value is fed back into the Tariff Agent as the deployed pricing intensity, closing the monitoring-to-pricing feedback loop. This is the learning mechanism in action — it identifies the optimal intensity from the evaluated options and confirms that a moderate, consumer-acceptable strength is optimal, with the optimum forming a

robust plateau rather than a knife-edge value. Aggressive pricing beyond this point offers no additional benefit in this network.

7.6 Consolidated monitoring results.

Metric	Result
Waiting Time Reduction (proxy)	−0.46%
Customer Response Rate	+0.16%
Off-Peak Uplift	+0.45%
Pricing Efficiency	₹4.92 vs ₹15.00 (−0.56%)
Optimal pricing strength (learned)	50

7.7 Conclusion. The Monitoring Agent confirms that dynamic pricing in this network substantially reduces the rare congestion (in relative terms), shifts a modest amount of demand into off-peak hours, and operates at essentially revenue-neutral pricing efficiency (−0.56%). The learning loop shows that a moderate pricing intensity (0.6) is optimal — neither passive nor aggressive — and that the optimum is a robust plateau rather than a knife-edge value. Taken together with Step 6, the consistent conclusion is that dynamic pricing here functions as a demand-balancing and congestion-easing tool operating at near-revenue-neutrality, and that conservative pricing is not only consumer-friendly but also reward-optimal.

Assumptions and limitations. The wait-time metric is a congestion proxy, not a measured queue length, as the data contains no direct waiting-time field; results indicate relative change in saturation rather than literal minutes saved. The baseline congestion rate is measured on predicted (smoothed) utilization, so it understates the actual saturation rate. The learning loop is a simulated feedback mechanism operating over a fixed historical dataset using the elasticity assumption; it demonstrates the mechanism of self-improvement and optimum-selection rather than learning from genuinely new live operational outcomes.

Step 8: Consolidated Evaluation

This section brings together the performance metrics from all three agents into a single evaluation scorecard, measured against the project's required metrics and the flat ₹15/kWh baseline. All pricing and monitoring figures reflect the learned-optimal pricing strength of 0.6.

8.1 Demand Prediction Agent.

Metric	Result	Baseline
Utilization RMSE	0.391	Naive persistence: 0.0545
Utilization MAE	0.236	
Utilization R^2	0.950	Naive persistence: 0.902
Congestion ROC-AUC	0.997	
Charging Load R^2	0.974	

The selected XGBoost model predicts utilization with R^2 0.95, improving on a naive persistence baseline by approximately 28% in RMSE, confirming it captures genuine demand dynamics rather than simple autocorrelation. Congestion classification and charging-load prediction also performed strongly.

8.2 Tariff Pricing Agent. Pricing performance was evaluated on ACN sessions against the ₹15/kWh baseline.

Metric	Result
Revenue Gain % (smooth, elasticity -0.3)	0.40%
Revenue Gain % (threshold, elasticity -0.3)	0.96%
Revenue range across elasticities (smooth)	0.31% to -0.45%
Revenue range across price bounds (± 10 –50%)	0.38% to -0.84%

The optimised smooth strategy consistently outperformed the rule-based threshold strategy. Revenue remained very close to baseline (within $\sim 0.5\%$) and was insensitive to the price bound, indicating dynamic pricing here is near-revenue-neutral and that wide, consumer-unfriendly price swings offer no revenue advantage.

8.3 Monitoring & Learning Agent.

metric	result
Awaiting Time Reduction (proxy)	-.46%
Customer Response Rate	+1.16%
Off-Peak Uplift	+0.45%
Pricing Efficiency (₹/kWh)	1.92 vs 15.00 (-0.56%)
Optimal pricing strength (learned)	50

Dynamic pricing substantially reduced the (rare) congestion, shifted a modest amount of demand into cheaper off-peak hours, and operated at near-baseline pricing efficiency. The learning mechanism selected a moderate pricing intensity (0.6) as reward-optimal.

8.4 Charger utilization: before vs after (peak-shaving effect). A before-and-after comparison of utilization confirmed the pricing's redistribution effect.

measure	before	after	change
Peak-hour utilization	340	330	-1.8%
Overall mean utilization	288	283	-1.7%

Utilization at peak hours fell noticeably more than the overall average, while demand shifted into the cheaper off-peak hours (+0.45%). This is a peak-shaving effect: the pricing suppresses demand most where it is highest and lifts it where it is lowest, flattening the demand curve. The slight decline in average utilization is the expected consequence of peak relief at a deliberately gentle, consumer-acceptable pricing intensity, not a loss of usage, and is consistent with the 54% relative congestion reduction reported by the Monitoring Agent.

8.5 Requirement coverage. All metrics specified in the project brief were computed and reported: RMSE, MAE, and R^2 for the Demand Prediction Agent; Revenue Gain %, Charger Utilization Rate (before/after), and Off-Peak Uplift for the Tariff Pricing Agent; and Average Waiting Time Reduction, Customer Response Rate, and

Pricing Efficiency Score for the Monitoring & Learning Agent. Off-peak uplift is operationalized as the demand share shifting into discounted hours, which correspond to the network's low-utilization periods.

8.6 Overall conclusion. The consolidated results tell a consistent story. The demand model forecasts utilization accurately (R^2 0.95). The pricing agent uses those forecasts to flatten the demand curve — shaving the peak by roughly 3%, easing the rare congestion by ~54% in relative terms, and gently lifting off-peak demand — all at near-revenue-neutrality (−0.4%). The learning agent confirms that a moderate, consumer-acceptable pricing intensity (0.6) is optimal. The unifying finding is that in this low-utilization, price-inelastic fleet network, dynamic pricing is most valuable as a demand-balancing and congestion-relief mechanism rather than a revenue-maximisation tool, and that conservative pricing is both consumer-friendly and reward-optimal.